CSA0905 JAVA PROGRAMMING

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EMPLOYEE.JAVA

PACKAGE APPLICATION

```
package Application;

public class Employee {

    private int employeeId;
    private String name;
    private String gender;
    private String dob;
    private String contact;
    private String email;
    private String address;
    private String department;
    private  String  designation;
    private String joiningDate;
    private double basicSalary;
    private double allowances;
    private double deductions;
    private String status;

    public Employee() {
    }

    public Employee(int employeeId, String name, String gender, String dob,
                    String contact, String email, String address,
                    String department, String designation, String joiningDate,
                    double basicSalary, double allowances,
                    double deductions, String status) {

        this.employeeId = employeeId;
        this.name = name;
        this.gender = gender;
```

```
        this.dob = dob;
        this.contact = contact;
        this.email = email;
        this.address = address;
        this.department = department;
        this.designation = designation;
        this.joiningDate = joiningDate;
        this.basicSalary = basicSalary;
        this.allowances = allowances;
        this.deductions = deductions;
        this.status = status;
    }

    public int getEmployeeId() {
        return employeeId;
    }

    public void setEmployeeId(int employeeId) {
        this.employeeId = employeeId;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getGender() {
        return gender;
    }

    public void setGender(String gender) {
        this.gender = gender;
    }

    public String getDob() {
        return dob;
    }

    public void setDob(String dob) {
        this.dob = dob;
```

```
}

public String getContact() {
    return contact;
}

public void setContact(String contact) {
    this.contact = contact;
}

public String getEmail() {
    return email;
}

public void setEmail(String email) {
    this.email = email;
}

public String getAddress() {
    return address;
}

public void setAddress(String address) {
    this.address = address;
}

public String getDepartment() {
    return department;
}

public void setDepartment(String department) {
    this.department = department;
}

public String getDesignation() {
    return designation;
}

public void setDesignation(String designation) {
    this.designation = designation;
}

public String getJoiningDate() {
```

```
        return joiningDate;
    }

    public void setJoiningDate(String joiningDate) {
        this.joiningDate = joiningDate;
    }

    public double getBasicSalary() {
        return basicSalary;
    }

    public void setBasicSalary(double basicSalary) {
        this.basicSalary = basicSalary;
    }

    public double getAllowances() {
        return allowances;
    }

    public void setAllowances(double allowances) {
        this.allowances = allowances;
    }

    public double getDeductions() {
        return deductions;
    }

    public void setDeductions(double deductions) {
        this.deductions = deductions;
    }

    public String getStatus() {
        return status;
    }

    public void setStatus(String status) {
        this.status = status;
    }
}
```

DBCONNECTION.JAVA

```
package org.example;

import java.sql.Connection;

import java.sql.DriverManager;

import java.sql.SQLException;
```

```
public class DBConnection {

    public static Connection getConnection() {

        Connection con = null;

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/employee",
                    "root",
                    """
                            MySQL@123"""
            );

        } catch (ClassNotFoundException e) {
            throw new RuntimeException(e);

        } catch (SQLException e) {
            throw new RuntimeException(e);
        }

        return con;
    }
}
```

EMPLOYEEMANAGEMENT.JAVA

```
        import Application.Employee;

import java.awt.*;
import java.awt.event.*;
import java.sql.ResultSet;
```

```
public class EmployeeManagement extends Frame {

    Label lblTitle, lblId, lblName, lblGender, lblDob, lblContact;
    Label lblEmail, lblAddress, lblDept, lblDesignation, lblJoining;
    Label lblBasic, lblAllowance, lblDeduction, lblStatus;

    TextField txtId, txtName, txtDob, txtContact, txtEmail;
    TextField txtAddress, txtDept, txtDesignation, txtJoining;
```

```
TextField txtBasic, txtAllowance, txtDeduction;

    Choice genderChoice;
    Choice statusChoice;

    Button btnAdd, btnSearch, btnUpdate, btnDelete;
    Button btnClear, btnView, btnPayroll;

    TextArea txtArea;

    public EmployeeManagement() {

        setTitle("Employee Management and Payroll Application");
        setSize(1000, 700);
        setLayout(new BorderLayout(10, 10));

        lblTitle = new Label("EMPLOYEE MANAGEMENT AND PAYROLL
SYSTEM", Label.CENTER);
        lblTitle.setFont(new Font("Arial", Font.BOLD, 22));
        add(lblTitle, BorderLayout.NORTH);

        Panel formPanel = new Panel();
        formPanel.setLayout(new GridLayout(8, 4, 8, 8));

        lblId = new Label("Employee ID");
        txtId = new TextField();

        lblName = new Label("Name");
        txtName = new TextField();

        lblGender = new Label("Gender");
        genderChoice = new Choice();
        genderChoice.add("Male");
        genderChoice.add("Female");

        lblDob = new Label("Date of Birth");
        txtDob = new TextField();

        lblContact = new Label("Contact Number");
        txtContact = new TextField();

        lblEmail = new Label("Email");
        txtEmail = new TextField();
```

```
lblAddress = new Label("Address");
txtAddress = new TextField();

lblDept = new Label("Department");
txtDept = new TextField();

lblDesignation = new Label("Designation");
txtDesignation = new TextField();

lblJoining = new Label("Date of Joining");
txtJoining = new TextField();

lblBasic = new Label("Basic Salary");
txtBasic = new TextField();

lblAllowance = new Label("Allowances");
txtAllowance = new TextField();

lblDeduction = new Label("Deductions");
txtDeduction = new TextField();

lblStatus = new Label("Employment Status");
statusChoice = new Choice();
statusChoice.add("Active");
statusChoice.add("Inactive");

formPanel.add(lblId);
formPanel.add(txtId);
formPanel.add(lblName);
formPanel.add(txtName);

formPanel.add(lblGender);
formPanel.add(genderChoice);
formPanel.add(lblDob);
formPanel.add(txtDob);

formPanel.add(lblContact);
formPanel.add(txtContact);
formPanel.add(lblEmail);
formPanel.add(txtEmail);

formPanel.add(lblAddress);
```

```
formPanel.add(txtAddress);
formPanel.add(lblDept);
formPanel.add(txtDept);

formPanel.add(lblDesignation);
formPanel.add(txtDesignation);
formPanel.add(lblJoining);
formPanel.add(txtJoining);

formPanel.add(lblBasic);
formPanel.add(txtBasic);
formPanel.add(lblAllowance);
formPanel.add(txtAllowance);

formPanel.add(lblDeduction);
formPanel.add(txtDeduction);
formPanel.add(lblStatus);
formPanel.add(statusChoice);

Panel buttonPanel = new Panel();
buttonPanel.setLayout(new FlowLayout());

btnAdd = new Button("Add");
btnSearch = new Button("Search");
btnUpdate = new Button("Update");
btnDelete = new Button("Delete");
btnClear = new Button("Clear");
btnView = new Button("View All");
btnPayroll = new Button("Payroll");

buttonPanel.add(btnAdd);
buttonPanel.add(btnSearch);
buttonPanel.add(btnUpdate);
buttonPanel.add(btnDelete);
buttonPanel.add(btnClear);
buttonPanel.add(btnView);
buttonPanel.add(btnPayroll);

Panel centerPanel = new Panel();
centerPanel.setLayout(new BorderLayout(10, 10));

centerPanel.add(formPanel, BorderLayout.NORTH);
centerPanel.add(buttonPanel, BorderLayout.CENTER);
```

```
add(centerPanel, BorderLayout.CENTER);

txtArea = new TextArea();
txtArea.setEditable(false);
add(txtArea, BorderLayout.SOUTH);

addWindowListener(new WindowAdapter() {
   public void windowClosing(WindowEvent e) {
      dispose();
   }
});

btnAdd.addActionListener(new ActionListener() {
   public void actionPerformed(ActionEvent e) {

      try {

         if (txtId.getText().isEmpty() ||
               txtName.getText().isEmpty() ||
               txtDob.getText().isEmpty() ||
               txtContact.getText().isEmpty() ||
               txtEmail.getText().isEmpty() ||
               txtAddress.getText().isEmpty() ||
               txtDept.getText().isEmpty() ||
               txtDesignation.getText().isEmpty() ||
               txtJoining.getText().isEmpty() ||
               txtBasic.getText().isEmpty() ||
               txtAllowance.getText().isEmpty() ||
               txtDeduction.getText().isEmpty()) {

            txtArea.setText("Please fill all mandatory fields.");
            return;
         }

         int id = Integer.parseInt(txtId.getText());
         double basic = Double.parseDouble(txtBasic.getText());
         double allowance = Double.parseDouble(txtAllowance.getText());
         double deduction = Double.parseDouble(txtDeduction.getText());

         if (basic < 0 || allowance < 0 || deduction < 0) {
            txtArea.setText("Salary values cannot be negative.");
            return;
```

```
            }

            Employee employee = new Employee(
                    id,
                    txtName.getText(),
                    genderChoice.getSelectedItem(),
                    txtDob.getText(),
                    txtContact.getText(),
                    txtEmail.getText(),
                    txtAddress.getText(),
                    txtDept.getText(),
                    txtDesignation.getText(),
                    txtJoining.getText(),
                    basic,
                    allowance,
                    deduction,
                    statusChoice.getSelectedItem()
            );

            EmployeeDAO dao = new EmployeeDAO();

            boolean result = dao.insertEmployee(employee);

            if (result) {
                txtArea.setText("Employee added successfully!");
            } else {
                txtArea.setText("Failed to add employee.");
            }

        } catch (NumberFormatException ex) {
            txtArea.setText("Invalid numeric input.");
        } catch (Exception ex) {
            txtArea.setText("Error: " + ex.getMessage());
        }
    }
});

btnSearch.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {

        try {

            int id = Integer.parseInt(txtId.getText());
```

```
                EmployeeDAO dao = new EmployeeDAO();
                ResultSet rs = dao.searchEmployee(id);

                if (rs.next()) {

                    txtName.setText(rs.getString("name"));
                    genderChoice.select(rs.getString("gender"));
                    txtDob.setText(rs.getString("dob"));
                    txtContact.setText(rs.getString("contact"));
                    txtEmail.setText(rs.getString("email"));
                    txtAddress.setText(rs.getString("address"));
                    txtDept.setText(rs.getString("department"));
                    txtDesignation.setText(rs.getString("designation"));
                    txtJoining.setText(rs.getString("joining_date"));
                    txtBasic.setText(String.valueOf(rs.getDouble("basic_salary")));

txtAllowance.setText(String.valueOf(rs.getDouble("allowances")));

txtDeduction.setText(String.valueOf(rs.getDouble("deductions")));
                    statusChoice.select(rs.getString("status"));

                    txtArea.setText("Employee found successfully.");

                } else {

                    txtArea.setText("Employee not found.");
                }

            } catch (NumberFormatException ex) {

                txtArea.setText("Enter a valid Employee ID.");

            } catch (Exception ex) {

                txtArea.setText("Error: " + ex.getMessage());
            }
        }
    });

    btnUpdate.addActionListener(new ActionListener() {
        public void actionPerformed(ActionEvent e) {
```

```
        try {

            int id = Integer.parseInt(txtId.getText());

            double basic = Double.parseDouble(txtBasic.getText());
            double allowance = Double.parseDouble(txtAllowance.getText());
            double deduction = Double.parseDouble(txtDeduction.getText());

            Employee employee = new Employee(
                    id,
                    txtName.getText(),
                    genderChoice.getSelectedItem(),
                    txtDob.getText(),
                    txtContact.getText(),
                    txtEmail.getText(),
                    txtAddress.getText(),
                    txtDept.getText(),
                    txtDesignation.getText(),
                    txtJoining.getText(),
                    basic,
                    allowance,
                    deduction,
                    statusChoice.getSelectedItem()
            );

            EmployeeDAO dao = new EmployeeDAO();

            boolean result = dao.updateEmployee(employee);

            if (result) {
                txtArea.setText("Employee updated successfully!");
            } else {
                txtArea.setText("Employee not found.");
            }

        } catch (Exception ex) {

            txtArea.setText("Error: " + ex.getMessage());
        }
    }
});

btnDelete.addActionListener(new ActionListener() {
```

```
    public void actionPerformed(ActionEvent e) {

        try {

            int id = Integer.parseInt(txtId.getText());

            EmployeeDAO dao = new EmployeeDAO();

            boolean result = dao.deleteEmployee(id);

            if (result) {

                txtArea.setText("Employee deleted successfully!");

            } else {

                txtArea.setText("Employee not found.");
            }

        } catch (Exception ex) {

            txtArea.setText("Error: " + ex.getMessage());
        }
    }
});

btnClear.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {

        txtId.setText("");
        txtName.setText("");
        txtDob.setText("");
        txtContact.setText("");
        txtEmail.setText("");
        txtAddress.setText("");
        txtDept.setText("");
        txtDesignation.setText("");
        txtJoining.setText("");
        txtBasic.setText("");
        txtAllowance.setText("");
        txtDeduction.setText("");

        genderChoice.select(0);
```

```
        statusChoice.select(0);

        txtArea.setText("");
    }
});

btnView.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {

        try {

            EmployeeDAO dao = new EmployeeDAO();
            ResultSet rs = dao.viewAll();

            StringBuilder result = new StringBuilder();

            result.append("EMPLOYEE DETAILS\n\n");

            while (rs.next()) {

                result.append("ID: ")
                      .append(rs.getInt("employee_id"))
                      .append("\n");

                result.append("Name: ")
                      .append(rs.getString("name"))
                      .append("\n");

                result.append("Gender: ")
                      .append(rs.getString("gender"))
                      .append("\n");

                result.append("Department: ")
                      .append(rs.getString("department"))
                      .append("\n");

                result.append("Designation: ")
                      .append(rs.getString("designation"))
                      .append("\n");

                result.append("Basic Salary: ")
                      .append(rs.getDouble("basic_salary"))
                      .append("\n");
```

```
                result.append("Allowances: ")
                      .append(rs.getDouble("allowances"))
                      .append("\n");

                result.append("Deductions: ")
                      .append(rs.getDouble("deductions"))
                      .append("\n");

                result.append("Status: ")
                      .append(rs.getString("status"))
                      .append("\n");

                result.append("----------\n");
            }

            txtArea.setText(result.toString());

        } catch (Exception ex) {

            txtArea.setText("Error: " + ex.getMessage());
        }
    }
});

btnPayroll.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {

        try {

            double basic = Double.parseDouble(txtBasic.getText());
            double allowance = Double.parseDouble(txtAllowance.getText());
            double deduction = Double.parseDouble(txtDeduction.getText());

            double grossSalary = basic + allowance;
            double netSalary = grossSalary - deduction;

            txtArea.setText(
                  "PAYROLL DETAILS\n\n" +
                        "Basic Salary : " + basic + "\n" +
                        "Allowances   : " + allowance + "\n" +
                        "Deductions   : " + deduction + "\n" +
                        "Gross Salary : " + grossSalary + "\n" +
```

```
                        "Net Salary   : " + netSalary
                );

            } catch (NumberFormatException ex) {

                txtArea.setText("Enter valid salary values.");
            }
        }
    });

    setVisible(true);
  }

  public static void main(String[] args) {
    new EmployeeManagement();
  }
}
```

EMPLOYEEDA0.JAVA

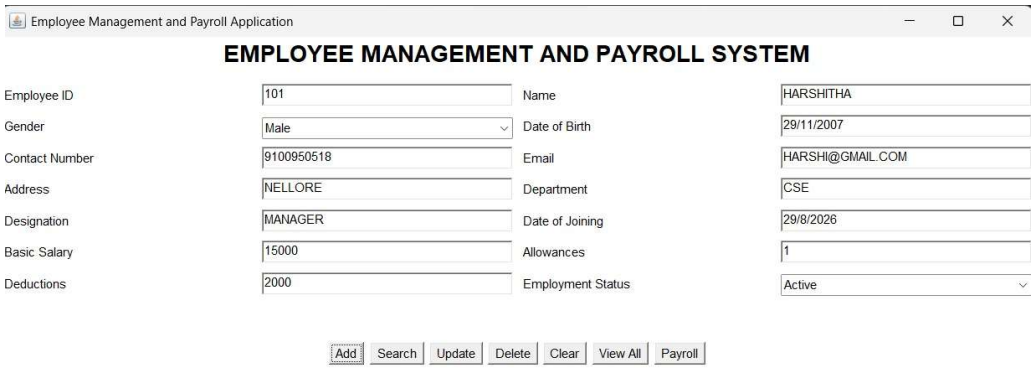
```
import Application.Employee;

import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;

public class EmployeeDAO {

  public boolean insertEmployee(Employee e) {

    String sql = "INSERT INTO employee
VALUES(?,?,?,?,?,?,?,?,?,?,?,?,?,?)";

    try {
      Connection con = DBConnection.getConnection();
      PreparedStatement ps = con.prepareStatement(sql);

      ps.setInt(1, e.getEmployeeId());
      ps.setString(2, e.getName());
      ps.setString(3, e.getGender());
      ps.setString(4, e.getDob());
      ps.setString(5, e.getContact());
      ps.setString(6, e.getEmail());
```

```
            ps.setString(7, e.getAddress());
            ps.setString(8, e.getDepartment());
            ps.setString(9, e.getDesignation());
            ps.setString(10, e.getJoiningDate());
            ps.setDouble(11, e.getBasicSalary());
            ps.setDouble(12, e.getAllowances());
            ps.setDouble(13, e.getDeductions());
            ps.setString(14, e.getStatus());

            int rows = ps.executeUpdate();

            ps.close();
            con.close();

            return rows > 0;

        } catch (SQLException ex) {
            throw new RuntimeException(ex);
        }
    }

    public ResultSet searchEmployee(int id) {

        String sql = "SELECT * FROM employee WHERE employee_id=?";

        try {
            Connection con = DBConnection.getConnection();
            PreparedStatement ps = con.prepareStatement(sql);

            ps.setInt(1, id);

            return ps.executeQuery();

        } catch (SQLException ex) {
            throw new RuntimeException(ex);
        }
    }

    public boolean updateEmployee(Employee e) {

        String sql = "UPDATE employee SET name=?, gender=?, dob=?,
contact=?, email=?, address=?, department=?, designation=?, joining_date=?,
basic_salary=?, allowances=?, deductions=?, status=? WHERE
```

```
employee_id=?";

        try {
            Connection con = DBConnection.getConnection();
            PreparedStatement ps = con.prepareStatement(sql);

            ps.setString(1, e.getName());
            ps.setString(2, e.getGender());
            ps.setString(3, e.getDob());
            ps.setString(4, e.getContact());
            ps.setString(5, e.getEmail());
            ps.setString(6, e.getAddress());
            ps.setString(7, e.getDepartment());
            ps.setString(8, e.getDesignation());
            ps.setString(9, e.getJoiningDate());
            ps.setDouble(10, e.getBasicSalary());
            ps.setDouble(11, e.getAllowances());
            ps.setDouble(12, e.getDeductions());
            ps.setString(13, e.getStatus());
            ps.setInt(14, e.getEmployeeId());

            int rows = ps.executeUpdate();

            ps.close();
            con.close();

            return rows > 0;

        } catch (SQLException ex) {
            throw new RuntimeException(ex);
        }
    }

    public boolean deleteEmployee(int id) {

        String sql = "DELETE FROM employee WHERE employee_id=?";

        try {
            Connection con = DBConnection.getConnection();
            PreparedStatement ps = con.prepareStatement(sql);

            ps.setInt(1, id);
```

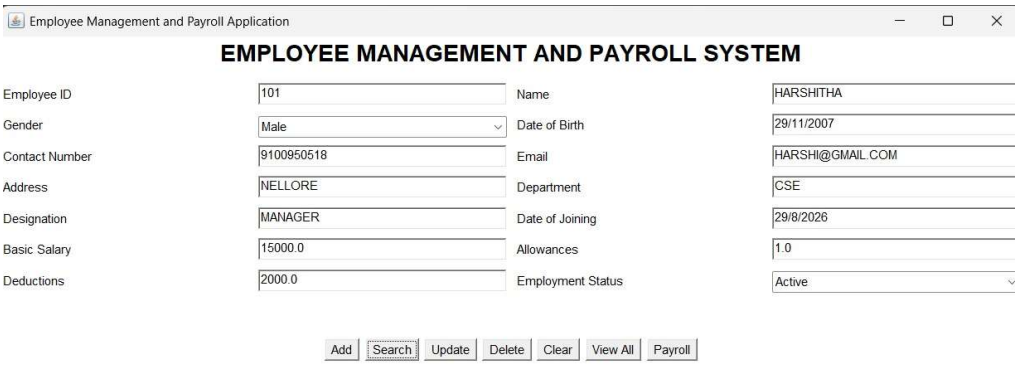
```
            int rows = ps.executeUpdate();

            ps.close();
            con.close();

            return rows > 0;

        } catch (SQLException ex) {
            throw new RuntimeException(ex);
        }
    }

    public ResultSet viewAll() {

        String sql = "SELECT * FROM employee";

        try {
            Connection con = DBConnection.getConnection();
            PreparedStatement ps = con.prepareStatement(sql);

            return ps.executeQuery();

        } catch (SQLException ex) {
            throw new RuntimeException(ex);
        }
    }
}
```

ADD -


Employee Management and Payroll Application
EMPLOYEE MANAGEMENT AND PAYROLL SYSTEM
Employee ID
101
Name
HARSHITHA
Gender
Male
Date of Birth
29/11/2007
Contact Number
9100950518
Email
HARSHI@GMAIL.COM
Address
NELLORE
Department
CSE
Designation
MANAGER
Date of Joining
29/8/2026
Basic Salary
15000
Allowances
1
Deductions
2000
Employment Status
Active
Add
Search
Update
Delete
Clear
View All
Payroll


Employee added successfully!

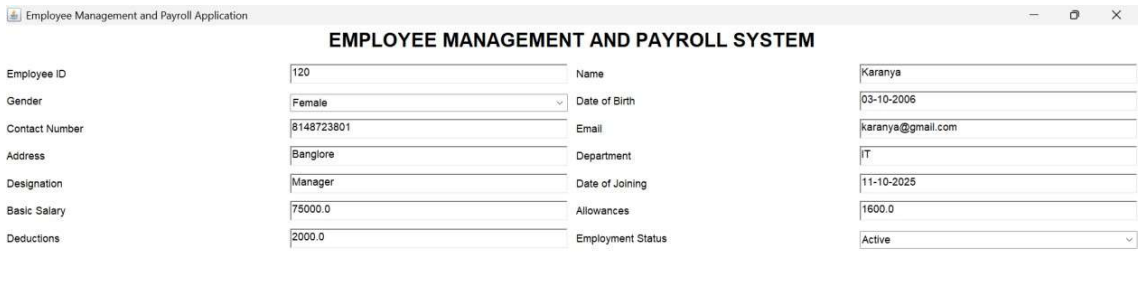
SEARCH

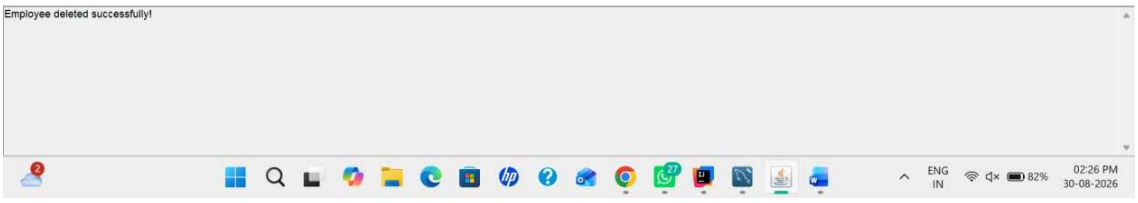

Employee Management and Payroll Application
EMPLOYEE MANAGEMENT AND PAYROLL SYSTEM
Employee ID
101
Name
HARSHITHA
Gender
Male
Date of Birth
29/11/2007
Contact Number
9100950518
Email
HARSHI@GMAIL.COM
Address
NELLORE
Department
CSE
Designation
MANAGER
Date of Joining
29/8/2026
Basic Salary
15000.0
Allowances
1.0
Deductions
2000.0
Employment Status
Active
Add
Search
Update
Delete
Clear
View All
Payroll


Employee found successfully!

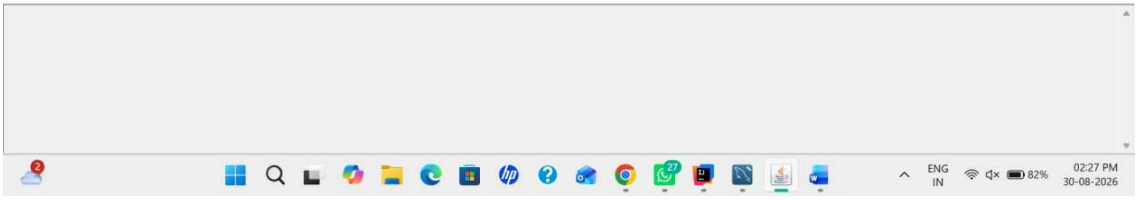
UPDATE

DELETE

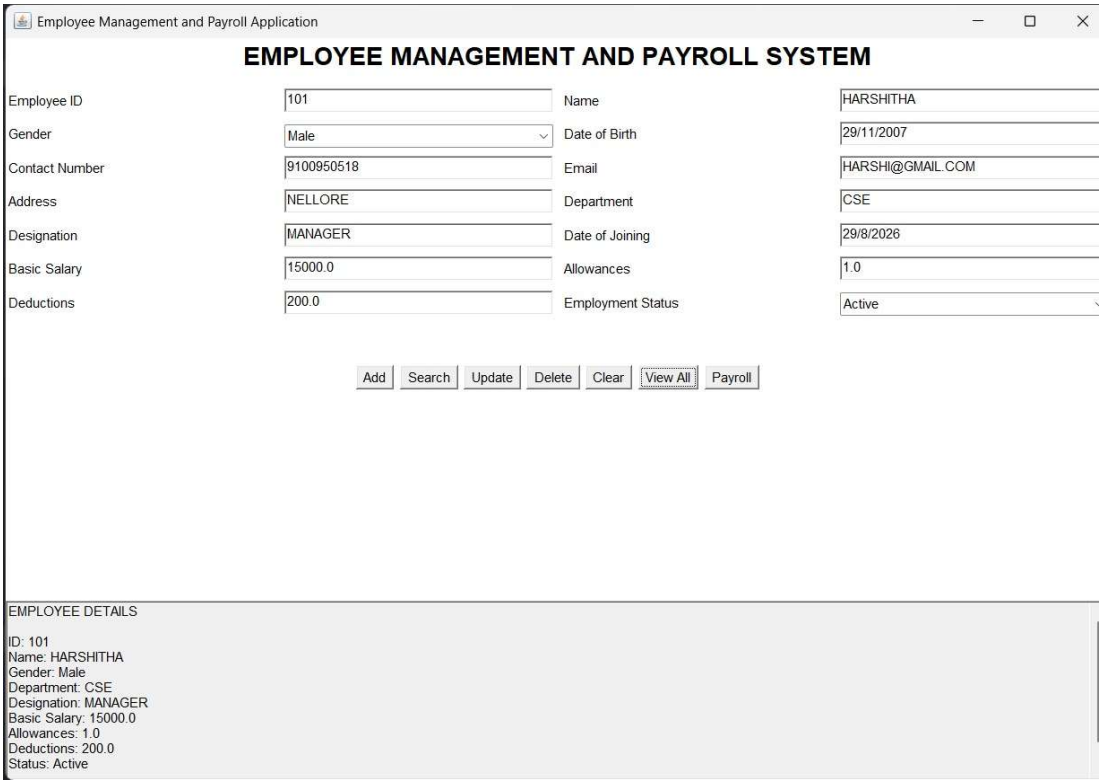




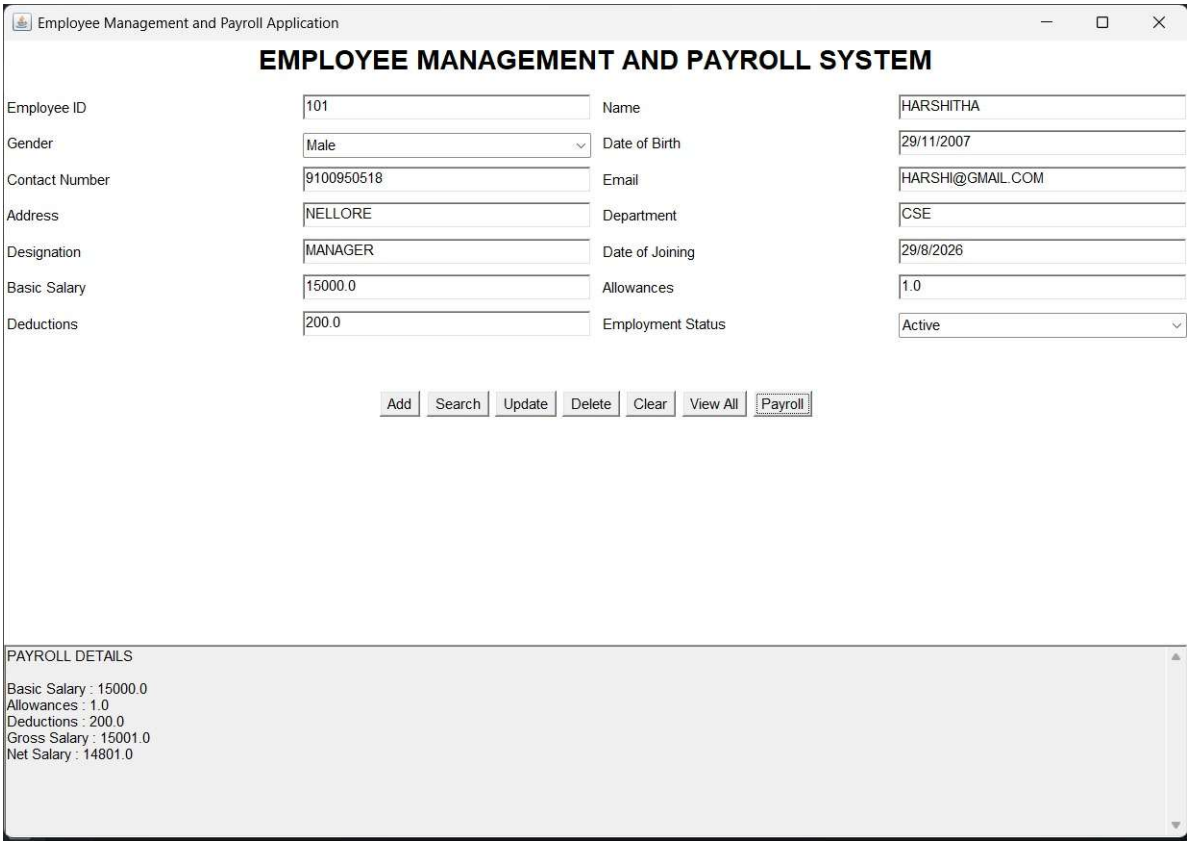
CLEAR



VIEWALL



PAYROLL


Employee Management and Payroll Application
EMPLOYEE MANAGEMENT AND PAYROLL SYSTEM
Employee ID
101
Name
HARSHITHA
Gender
Male
Date of Birth
29/11/2007
Contact Number
9100950518
Email
HARSHI@GMAIL.COM
Address
NELLORE
Department
CSE
Designation
MANAGER
Date of Joining
29/8/2026
Basic Salary
15000.0
Allowances
1.0
Deductions
200.0
Employment Status
Active
Add
Search
Update
Delete
Clear
View All
Payroll
PAYROLL DETAILS
Basic Salary : 15000.0
Allowances : 1.0
Deductions : 200.0
Gross Salary : 15001.0
Net Salary : 14801.0